



About Final Exam

Final Exam Arrangement

Info

- Time: July 27, 2026 (Monday), 9:30-11:30
- Location: TBD (currently being arranged, will be announced later on Canvas).

E4-102

帶 Student ID card, Pens

Exam Range:

- Contents in Lecture 1-Lecture 7

Question Type

- Single Choice Questions
- Short Answer Questions
- Code Understanding Questions (Derive the Output)
- Code Completion Questions

Simple Examples to Clarify Question Type

Single Choice Questions:

Which of the following statements about function is false?

- (A) You can assign functions to variables.
- (B) You can pass functions around and use them as arguments.
- (C) You can treat functions like any other object.
- (D) You can use functions only with specific data types.

Short Answer Questions:

List three unsupervised learning models supported by scikit-learn.

Answer: K-Means clustering, DBSCAN, Gaussian Mixtures.

Simple Examples to Clarify Question Type

Code Understanding Questions (Derive the Output)

```
def c(N, k):  
    if N < k:  
        return str(N)  
    else:  
        return c(N // k, k) + str(N % k)  
  
print(c(1234, 8))
```

answer:

2322 (5 points)

others (0 point)

Simple Examples to Clarify Question Type

Code Completion Questions

Complete a Python function which returns the **median** of a given list of numbers. Inside the function, you use `sorted()` to sort the samples in ascending order. Then you check if your list has an odd number of data points, in which case, you return the item in the middle directly. If the list has an even number of samples, then you compute the index of the two items in the middle, calculate their average, and return the resulting value.

```
def median(samples) :  
    n = len(samples)  
    middle_index = n // 2  
    sorted_samples = sorted(samples)  
    if n % 2 == 1 :  
        return -----  
    else :  
        return -----
```
